

Without Scaling:



STATISTICS

CHARTS

FAILURES

EXCEPTIONS

CURRENT RATIO

DOWNLOAD DATA

LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/products/search?q=alpha	7900	0	38	49	61	39.42	27	216	1094.98	21.2	0
GET	/products/search?q=beta	7752	0	38	49	58	39.3	25	211	1044.99	18.1	0
GET	/products/search?q=books	7960	0	38	48	59	39.25	24	211	1044.99	21.2	0
GET	/products/search?q=electronics	7813	0	38	49	60	39.26	25	212	1094.98	19.6	0
GET	/products/search?q=home	7825	0	38	49	59	39.36	27	205	974.99	21	0
GET	/products/search?q=product	7881	0	38	49	59	39.41	27	214	1971.98	18.1	0
	Aggregated	47131	0	38	49	59	39.33	24	216	1205.04	119.2	0

Health

Alarm recommendations

Investigate with AI - new

1h

3h

12h

1d

3d

1w

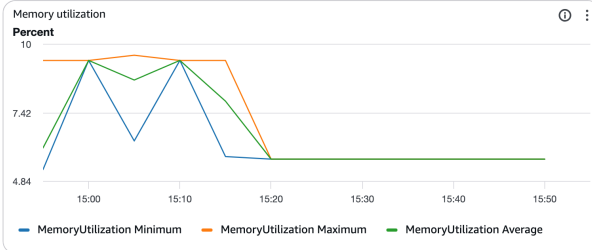
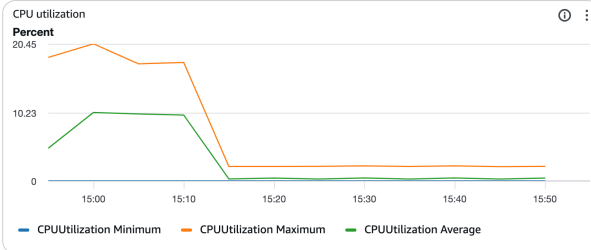
Custom

Local timezone

Refresh

Download

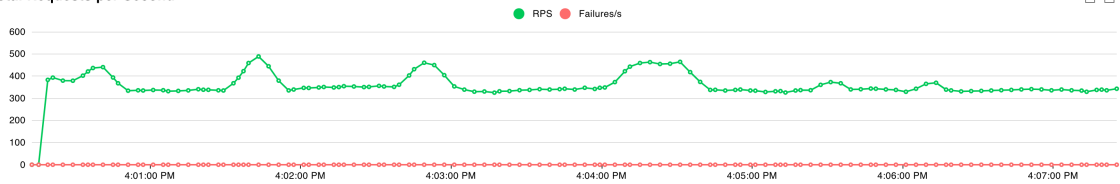
Explore related



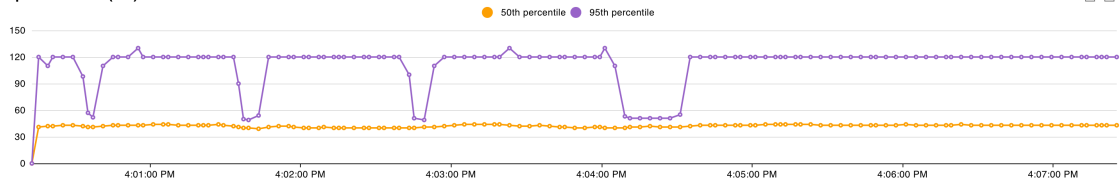
5-user ECS metrics

STATISTICS CHARTS FAILURES EXCEPTIONS CURRENT RATIO DOWNLOAD DATA LOGS

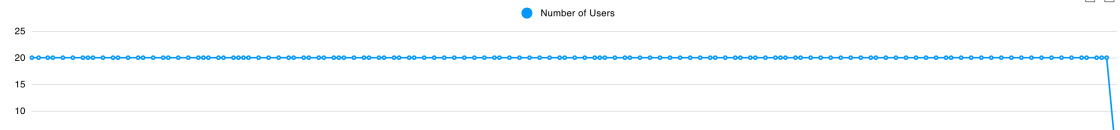
Total Requests per Second



Response Times (ms)



Number of Users



20-user chart

STATISTICS

CHARTS

FAILURES

EXCEPTIONS

CURRENT RATIO

DOWNLOAD DATA

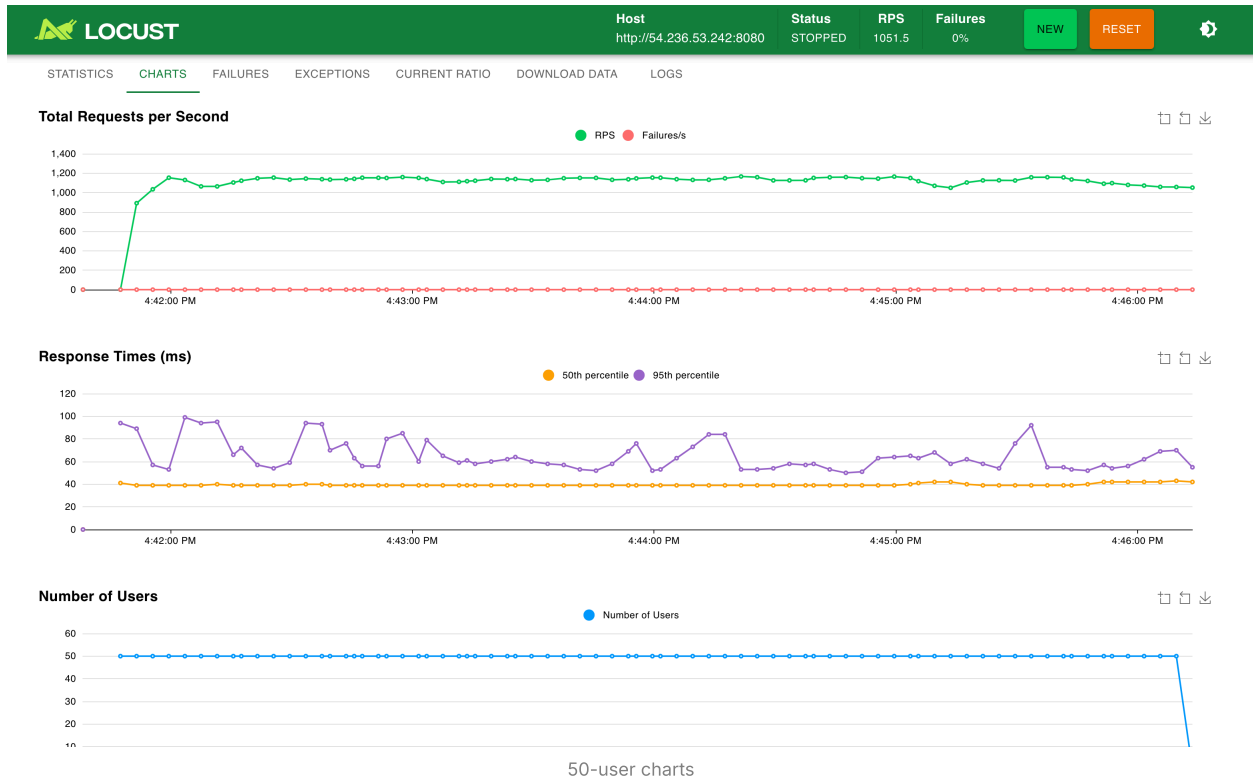
LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/products/search?q=alpha	25565	0	42	120	130	54.28	26	253	1094.96	58.7	0
GET	/products/search?q=beta	25845	0	42	120	130	54.18	26	283	1044.96	54.9	0
GET	/products/search?q=books	25867	0	42	120	130	54.43	25	284	1044.96	57.1	0
GET	/products/search?q=electronics	25817	0	42	120	130	53.94	26	282	1094.95	58.8	0
GET	/products/search?q=home	26032	0	42	120	130	54.21	25	282	974.96	56.9	0
GET	/products/search?q=product	25760	0	42	120	130	54.57	26	282	1971.95	56.4	0
	Aggregated	154886	0	42	120	130	54.27	25	284	1203.95	342.8	0

20-user statistics



20-user ECS metrics



LOCUST Host: http://54.236.53.242:8080 Status: STOPPED RPS: 1051.5 Failures: 0% [NEW] [RESET] [Settings]

STATISTICS CHARTS FAILURES EXCEPTIONS CURRENT RATIO DOWNLOAD DATA LOGS

☰

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/products/search?q=alpha	49781	0	39	60	160	43.38	24	273	1094.94	175.6	0
GET	/products/search?q=beta	49996	0	39	60	160	43.39	24	267	1044.94	178	0
GET	/products/search?q=books	49611	0	39	60	160	43.35	24	265	1044.94	170.4	0
GET	/products/search?q=electronics	49591	0	39	60	160	43.41	24	256	1094.94	173.9	0
GET	/products/search?q=home	49503	0	39	60	160	43.4	24	268	974.94	176	0
GET	/products/search?q=product	50184	0	40	60	160	43.57	24	254	1971.94	177.6	0
Aggregated		298666	0	39	60	160	43.42	24	273	1205.73	1051.5	0

50-user statistics

Health

Alarm recommendations

Investigate with AI - new

1h

3h

12h

1d

3d

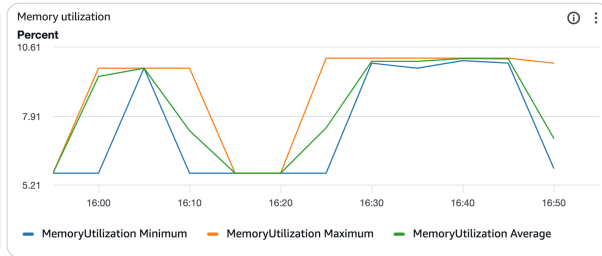
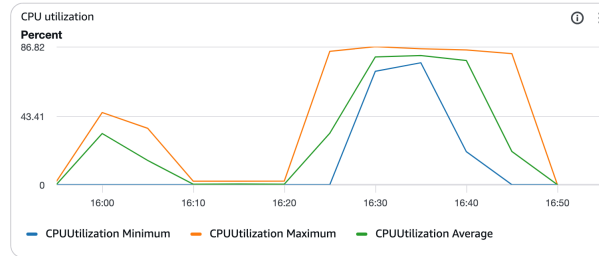
1w

Custom

Local timezone

Refresh

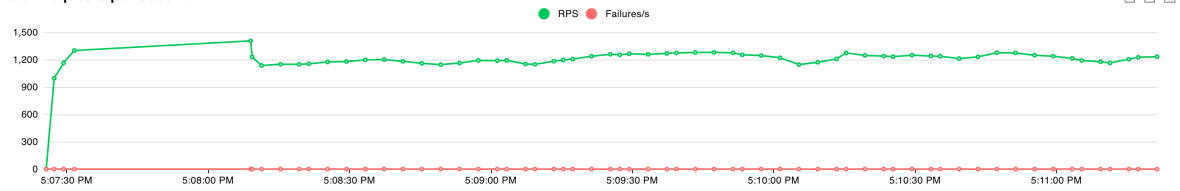
Explore related



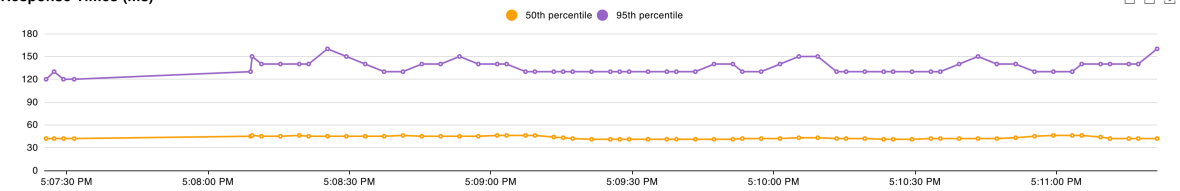
50-user ECS metrics

STATISTICS CHARTS FAILURES EXCEPTIONS CURRENT RATIO DOWNLOAD DATA LOGS

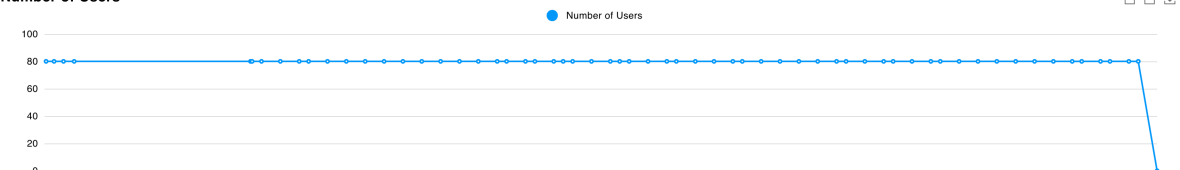
Total Requests per Second



Response Times (ms)



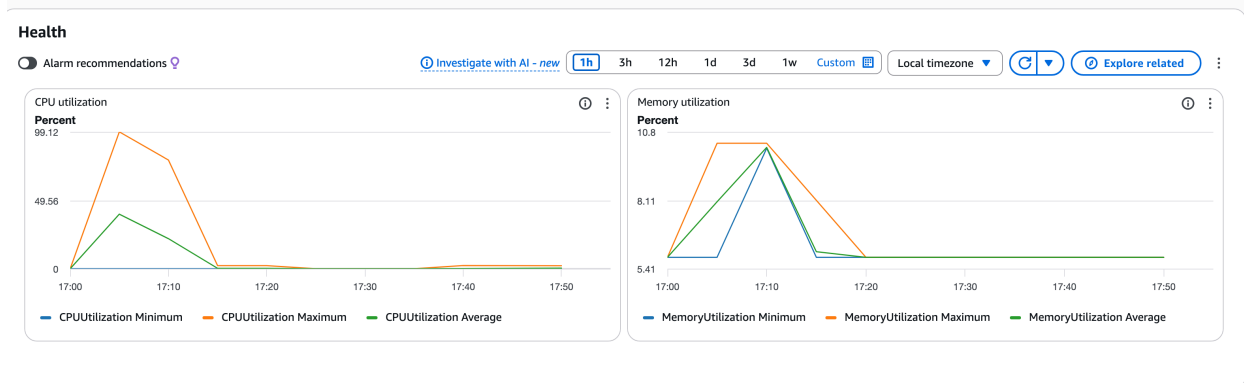
Number of Users



80-user ECS charts

STATISTICS												
Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/products/search?q=alpha	49933	0	43	130	200	61.94	25	352	1094.94	206.7	0
GET	/products/search?q=beta	50241	0	43	130	200	61.79	24	329	1044.94	204.6	0
GET	/products/search?q=books	49444	0	43	130	190	61.68	25	348	1044.94	199.6	0
GET	/products/search?q=electronics	49822	0	43	130	200	61.76	25	331	1094.93	199.4	0
GET	/products/search?q=home	50057	0	43	130	200	61.57	25	345	974.94	206.6	0
GET	/products/search?q=product	49696	0	43	130	200	61.78	25	326	1971.93	215.2	0
Aggregated		299193	0	43	130	200	61.75	24	352	1203.87	1232.1	0

80-user statistics



80-user ECS metrics

Users	Requests	Failures	RPS	Median (ms)	P95 (ms)	P99 (ms)	Avg (ms)
5	47,131	0	119.2	38	49	59	39.33
20	154,886	0	342.8	42	120	130	54.27
50	298,666	0	1051.5	39	60	160	43.42
80	299,193	0	1232.1	43	130	200	61.75

I initially ran the required 5-user and 20-user tests. The 20-user run increased CPU pressure and degraded latency, but it did not fully saturate the service. To further identify the bottleneck, I ran additional 50-user and 80-user tests. In the 80-user test, CloudWatch showed CPU utilization peaking at ~99% while memory stayed stable (~10–11%), and Locust showed worsening tail latency with only modest throughput gains relative to the increase in concurrency. This indicates the system is reaching its breaking point due to CPU/compute limits rather than memory pressure.

1. Which resource hits the limit first?

CPU

2. How much did response times degrade?

5 → 20:

Avg: 39.33 → 54.27 ms (+38%)

P95: 49 → 120 ms (+145%)

50 → 80:

Avg: 43.42 → 61.75 ms (+42%)

P95: 60 → 130 ms (+116.7%)

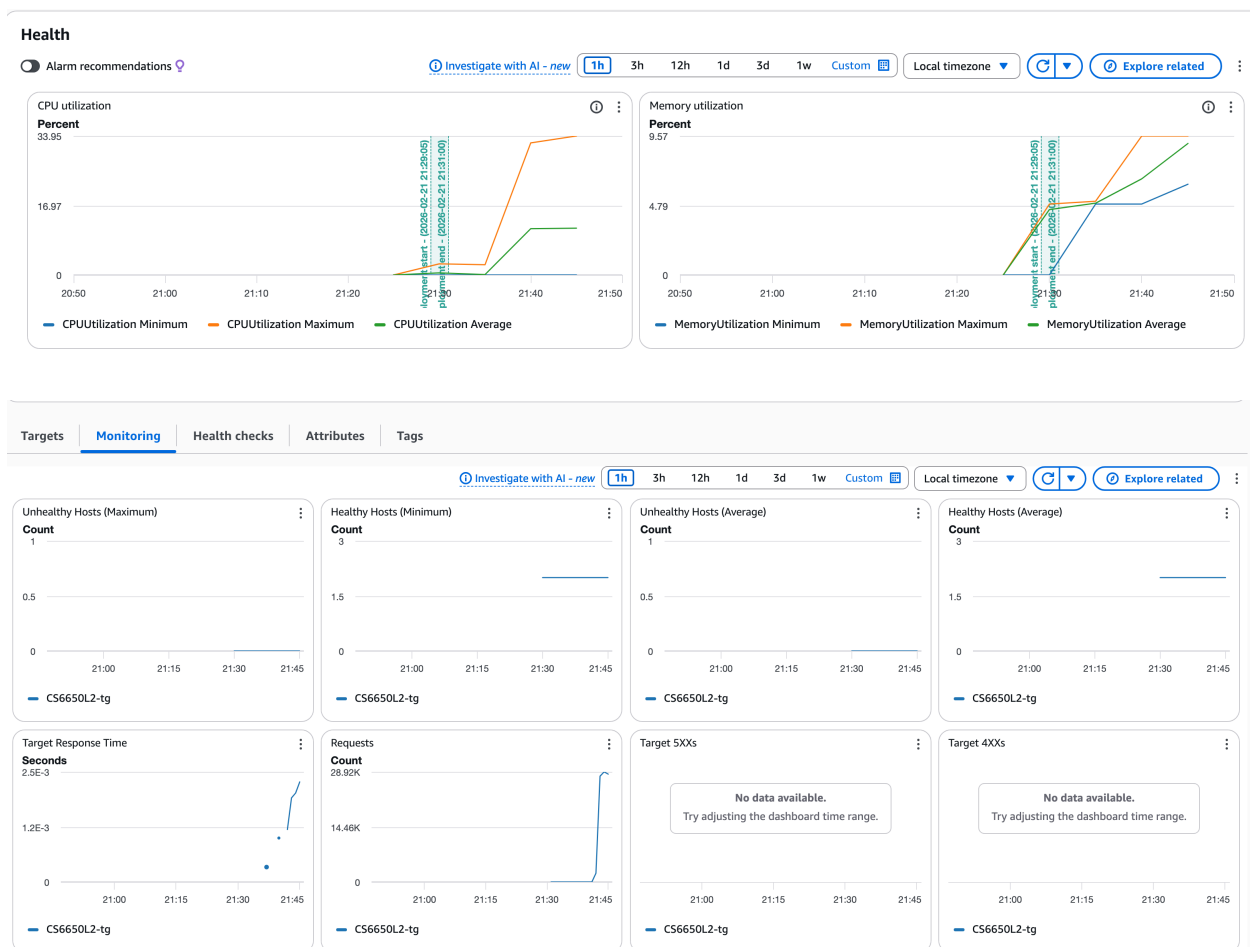
P99: 160 → 200 ms (+25%)

3. Could doubling CPU (256 → 512) help?

Yes, because the evidence shows CPU saturation while memory remains low/stable. Doubling CPU should improve latency and probably increase throughput before the next bottleneck is reached.

Auto Scaling

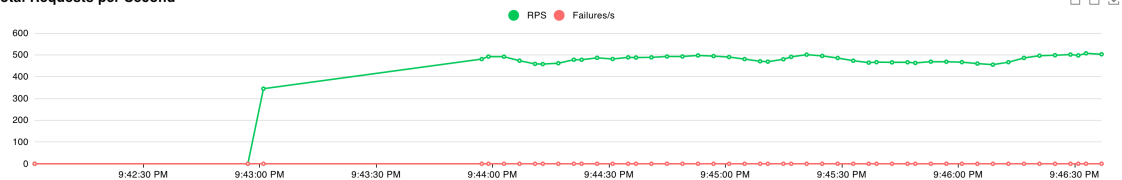
20 users:



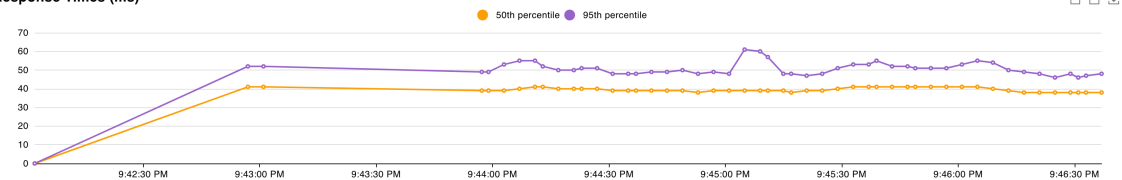


Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/products/search?q=alpha	17318	0	40	51	72	41.25	24	222	1095.01	85	0
GET	/products/search?q=beta	17444	0	40	52	79	41.42	25	235	1045.01	87.7	0
GET	/products/search?q=books	17539	0	40	52	77	41.27	25	229	1045.01	81.5	0
GET	/products/search?q=electronics	17593	0	40	52	73	41.27	25	245	1095.01	84.7	0
GET	/products/search?q=home	17494	0	40	51	74	41.25	25	227	975.01	82.4	0
GET	/products/search?q=product	17427	0	40	52	72	41.47	24	250	1972.01	81.1	0
Aggregated		104815	0	40	51	74	41.32	24	250	1204.11	502.4	0

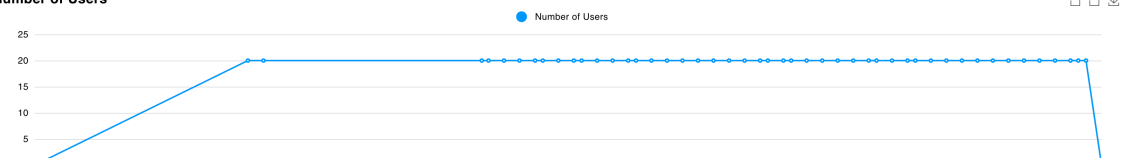
Total Requests per Second



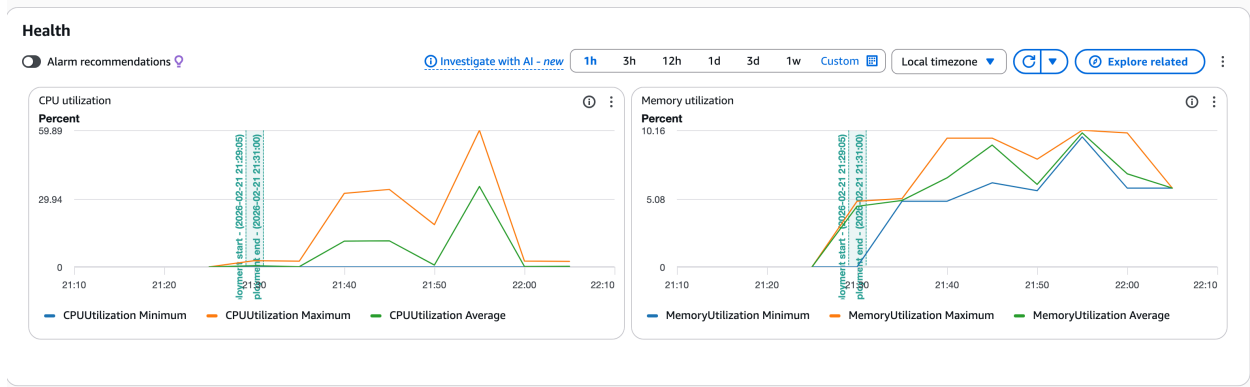
Response Times (ms)



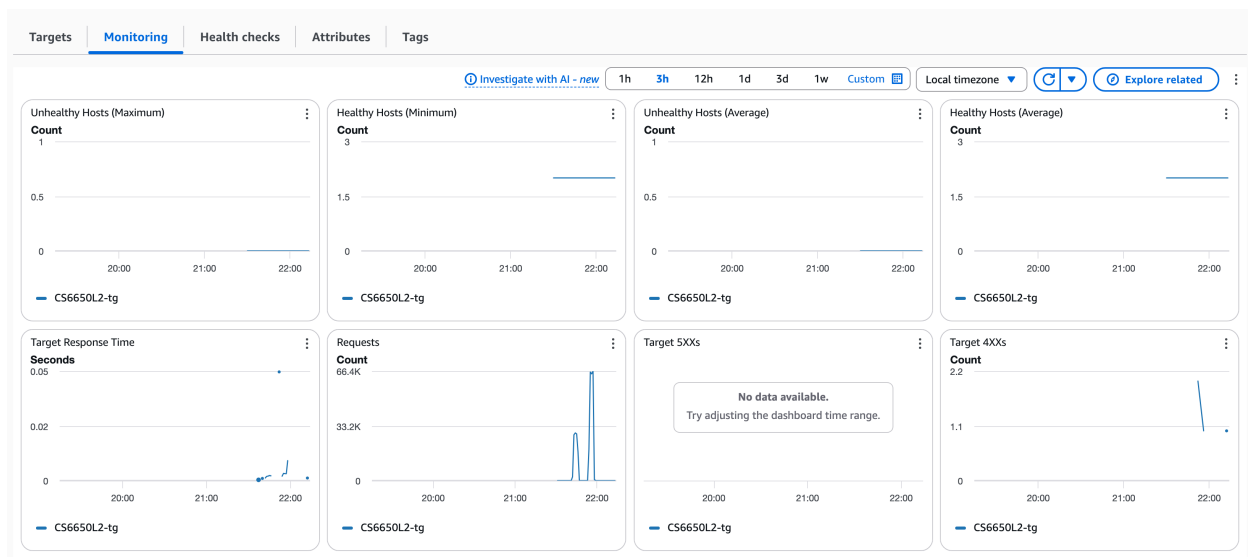
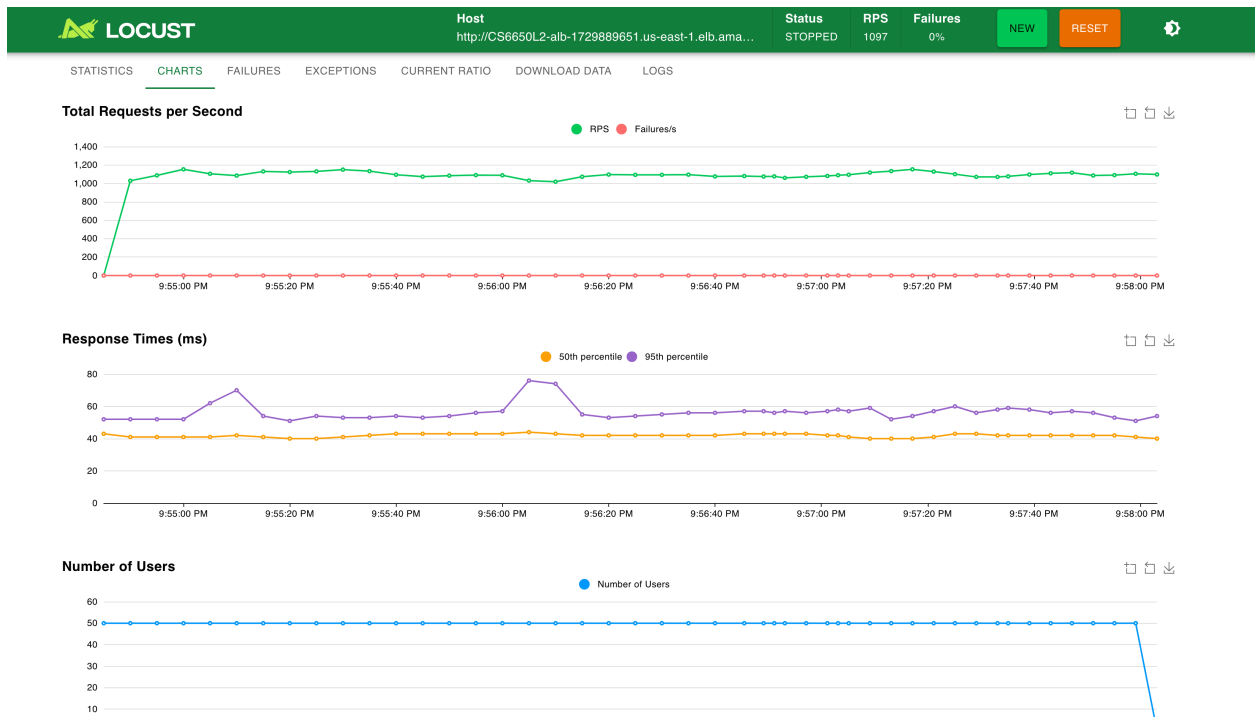
Number of Users



50 users:

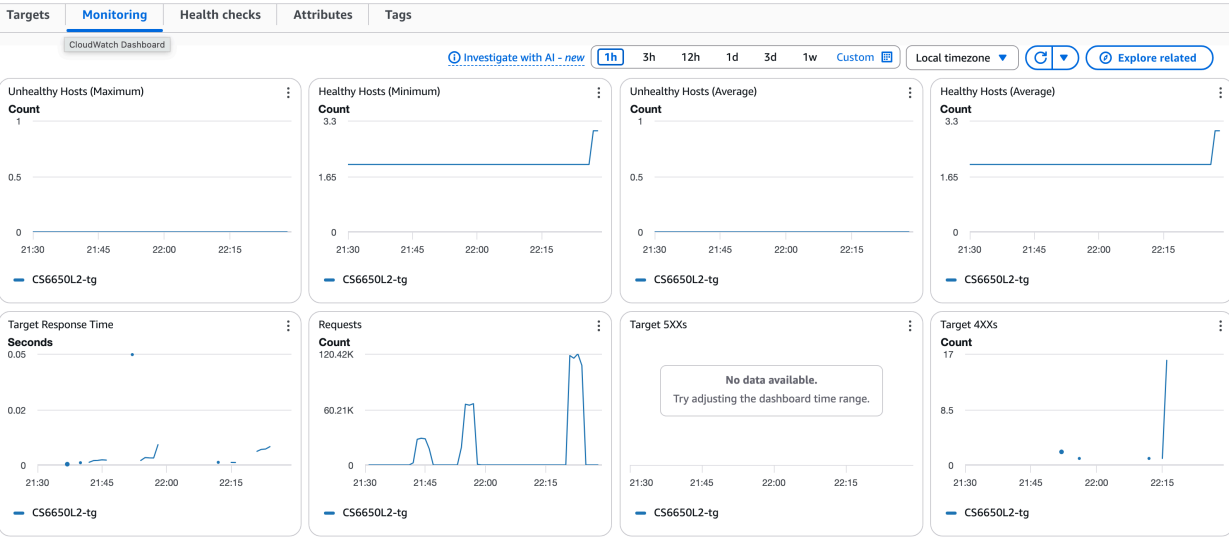
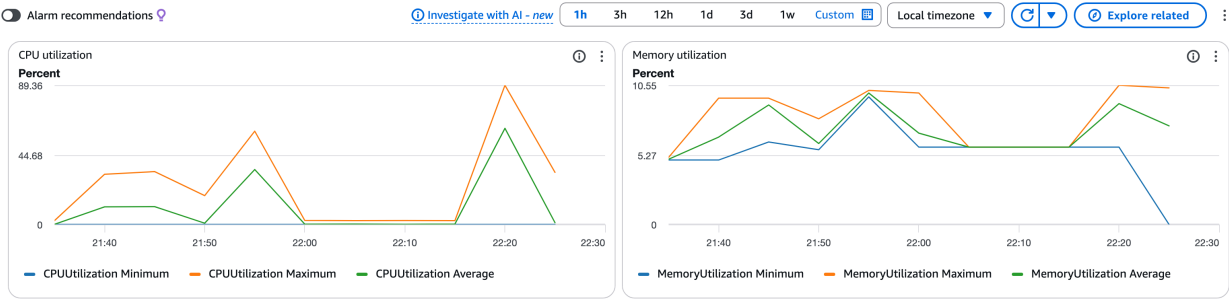


LOCUST Host http://CS6650L2-alb-1729889651.us-east-1.elb.ama... Status STOPPED RPS 1097 Failures 0% NEW RESET												
STATISTICS CHARTS FAILURES EXCEPTIONS CURRENT RATIO DOWNLOAD DATA LOGS												
Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/products/search?q=alpha	35979	0	42	57	130	44.74	25	294	1094.98	179.9	0
GET	/products/search?q=beta	36448	0	42	56	120	44.56	27	294	1044.98	191.9	0
GET	/products/search?q=books	35887	0	42	56	120	44.57	25	301	1044.98	175.2	0
GET	/products/search?q=electronics	36336	1	42	56	120	44.59	25	315	1094.94	182.5	0
GET	/products/search?q=home	35744	0	42	56	120	44.64	25	294	974.98	180.7	0
GET	/products/search?q=product	36238	0	42	56	120	44.68	27	316	1971.97	186.8	0
Aggregated		216632	1	42	56	120	44.63	25	316	1205.18	1097	0



100 users:

Health



LOCUST

Host
http://CS6650L2-alb-1729889651.us-east-1.elb.ama...

Status
STOPPED

RPS
2052

Failures
0%

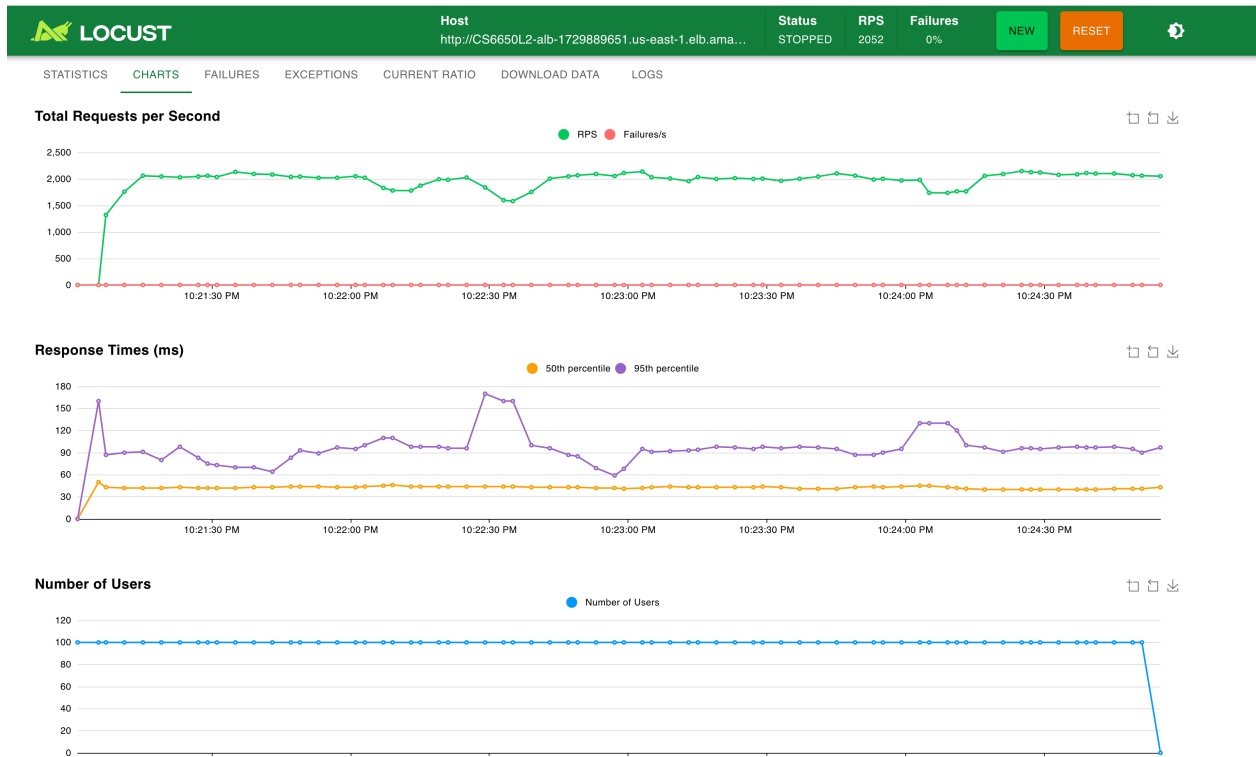
NEW

RESET

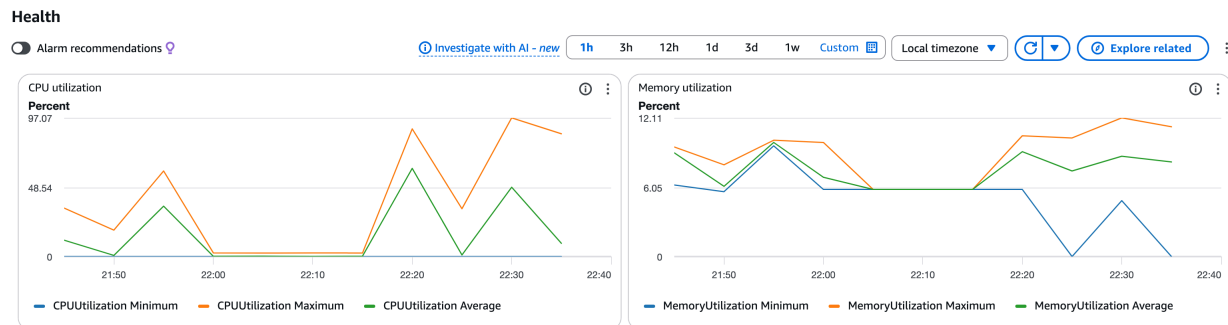
⚙️

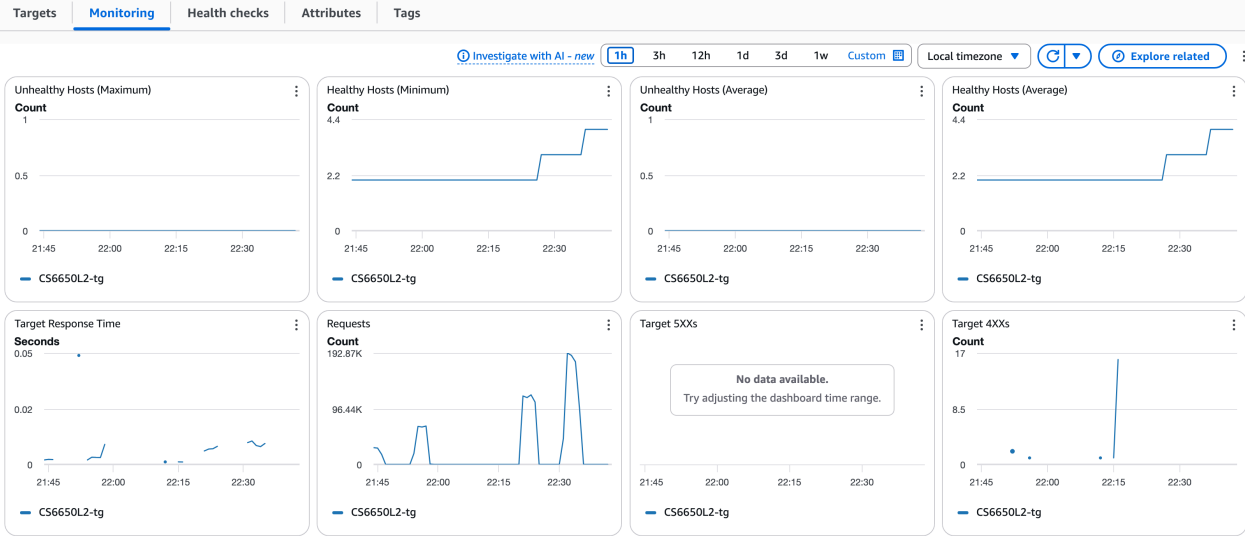
STATISTICS CHARTS FAILURES EXCEPTIONS CURRENT RATIO DOWNLOAD DATA LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/products/search?q=alpha	76850	0	42	97	180	49.01	24	450	1094.96	347.3	0
GET	/products/search?q=beta	77082	0	42	96	180	48.86	24	452	1044.96	339.8	0
GET	/products/search?q=books	77495	0	42	97	180	48.98	24	384	1044.96	350.2	0
GET	/products/search?q=electronics	76506	0	42	97	190	49.12	25	450	1094.96	336.8	0
GET	/products/search?q=home	76857	0	42	97	190	49.07	24	452	974.96	337.7	0
GET	/products/search?q=product	77587	0	43	97	180	49.15	25	401	1971.96	340.2	0
Aggregated		462377	0	42	97	180	49.03	24	452	1205.46	2052	0

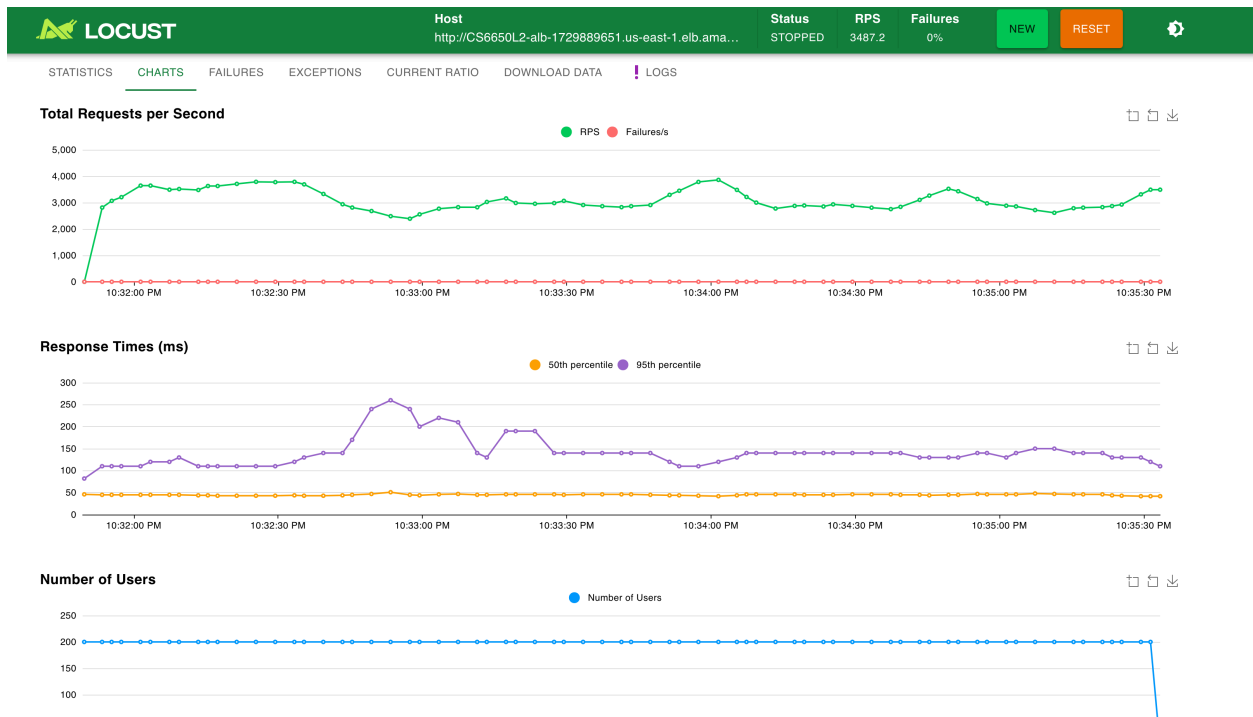


200 users:





 LOCUST Host http://CS6650L2-alb-1729889651.us-east-1.elb.ama... Status STOPPED RPS 3487.2 Failures 0% NEW RESET 												
STATISTICS CHARTS FAILURES EXCEPTIONS CURRENT RATIO DOWNLOAD DATA  LOGS												
Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/products/search?q=alpha	116505	0	45	140	230	62.03	24	491	1094.94	571.8	0
GET	/products/search?q=beta	116732	0	45	140	230	62.17	24	487	1044.95	592.1	0
GET	/products/search?q=books	117000	0	45	140	230	62.2	24	487	1044.94	590.1	0
GET	/products/search?q=electronics	116843	0	45	140	230	61.99	24	487	1094.94	575.4	0
GET	/products/search?q=home	116734	0	45	140	230	62.34	25	487	974.94	578.7	0
GET	/products/search?q=product	116963	0	45	140	230	62.06	26	486	1971.94	579.1	0
Aggregated		700777	0	45	140	230	62.13	24	491	1204.65	3487.2	0



Users	Total Requests	Median (ms)	P95 (ms)	P99 (ms)	Avg (ms)	ECS CPU Utilization	ECS Memory Utilization
20	104,815	40	51	74	41.32	~30–60% max (avg much lower)	~5–10%
50	216,632	42	56	120	44.63	~30–60% max (one spike)	~5–10%
100	462,377	42	97	180	49.03	~up to 80–90% max	~5–11%
200	700,777	45	140	230	62.13	~up to 90%+ max (avg elevated)	~5–12%

How does the system solved Part II bottleneck

With horizontal scaling enabled, the system handled the same type of load much better than a single-task setup. This improved capacity by adding **parallel compute resources**, which is exactly the right approach for a fixed-compute workload.

ALB (Application Load Balancer)

Routes traffic to backend ECS tasks and spreads load across healthy tasks instead of overloading one instance.

Target Group

Maintains the pool of backend task targets and runs **health checks** (e.g., `/health`) to decide which tasks are eligible to receive traffic. It can also provides useful monitoring metrics such as healthy hosts and response time

Auto Scaling (ECS Service Auto Scaling)

Monitors load and Automatically increases task count when CPU pressure is sustained.

Compare the trade-offs between this approach and vertical scaling

Horizontal Scaling

Pros

- Can scale up/down based on demand
- Can handle higher traffic by adding more tasks
- Better fault tolerance (one task failure does not take down the entire service)

Cons

- More infrastructure complexity (ALB, Target Group, health checks, scaling policy)
- More parts to monitor/debug

Vertical Scaling (bigger CPU/memory on one task)

Pros

- Simpler to configure and reason about
- Fast way to improve capacity for a single bottlenecked task

Cons

- Has an upper limit (cannot scale forever by increasing one task size)
- Single-task failure is still a single point of failure

Conclusion: For this CPU-heavy, fixed-workload search service, horizontal scaling is the better architectural solution than only increasing task size.

Predict scaling behavior for different load patterns

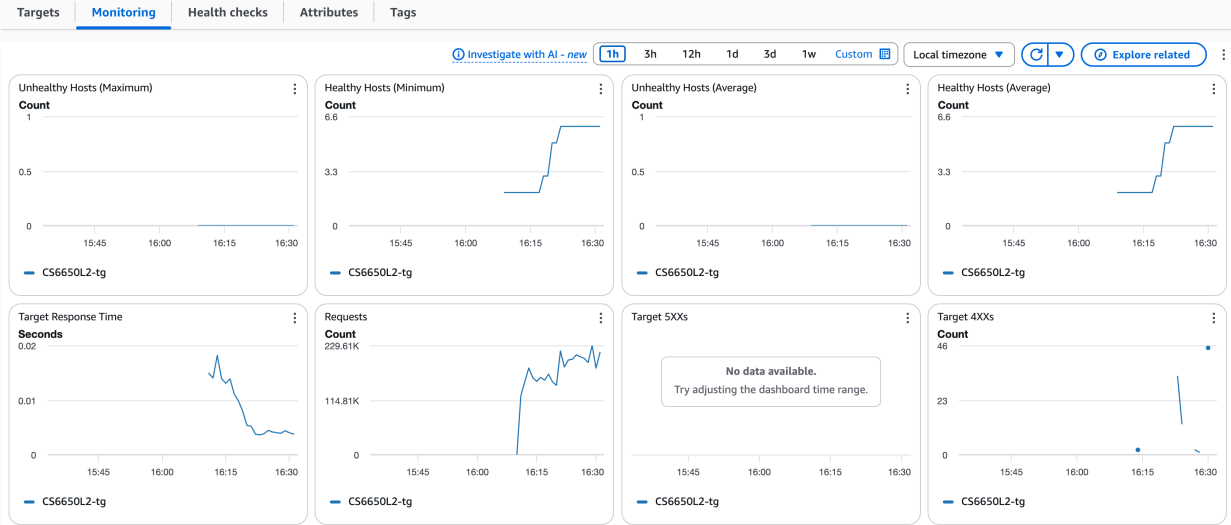
gradual increase: CPU rises progressively. Auto Scaling eventually triggers scale-out. Healthy Hosts increases step-by-step

sudden spike: Short-term latency spike is likely before new tasks are ready. CPU jumps first, then Auto Scaling reacts. After new tasks pass health checks, response time should improve or stabilize

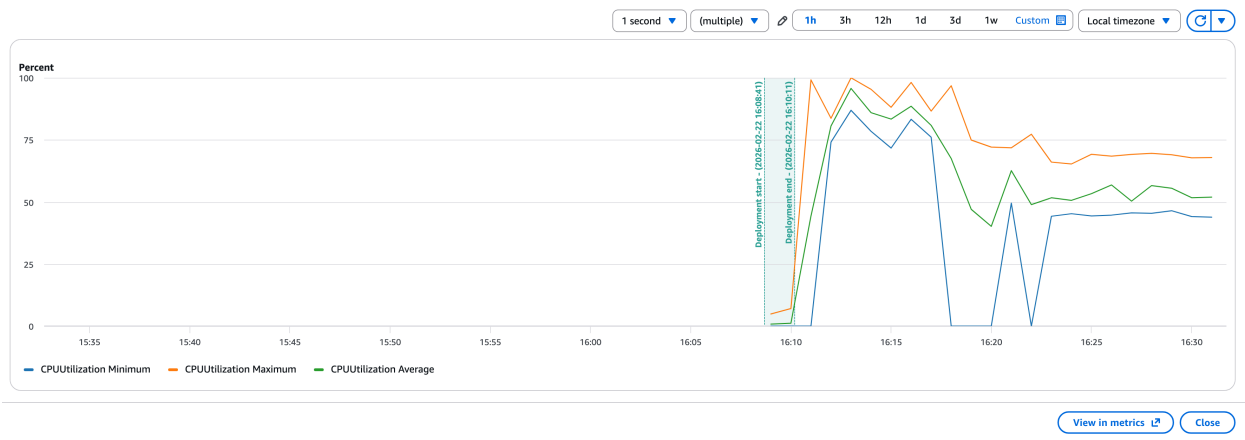
Variations

```
min_capacity = 2
max_capacity = 6
cpu_target_value = 60
scale_in_cooldown = 15
scale_out_cooldown = 15
```

200 users:



CPU utilization





LOCUST

Host

http://CS6650L2-alb-1187570867.us-east-1.elb.amaz...

Status

STOPPED

RPS

3059.6

Failures

0%

NEW

RESET



STATISTICS

CHARTS

FAILURES

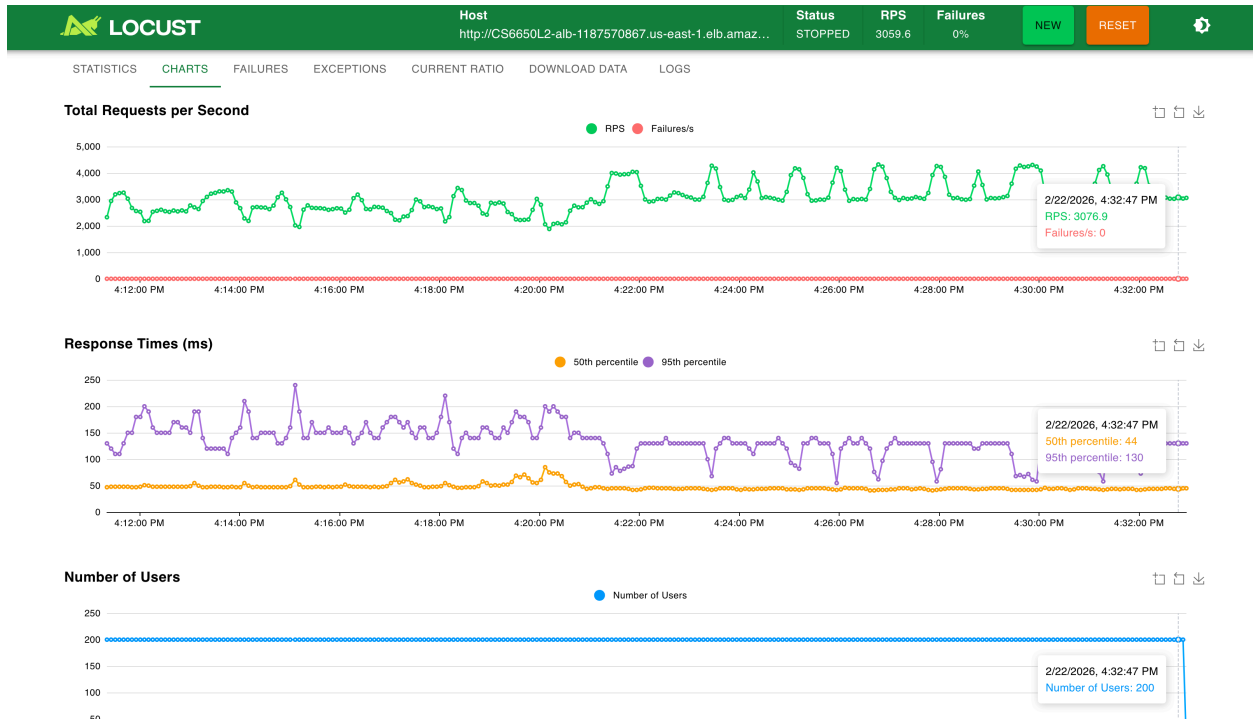
EXCEPTIONS

CURRENT RATIO

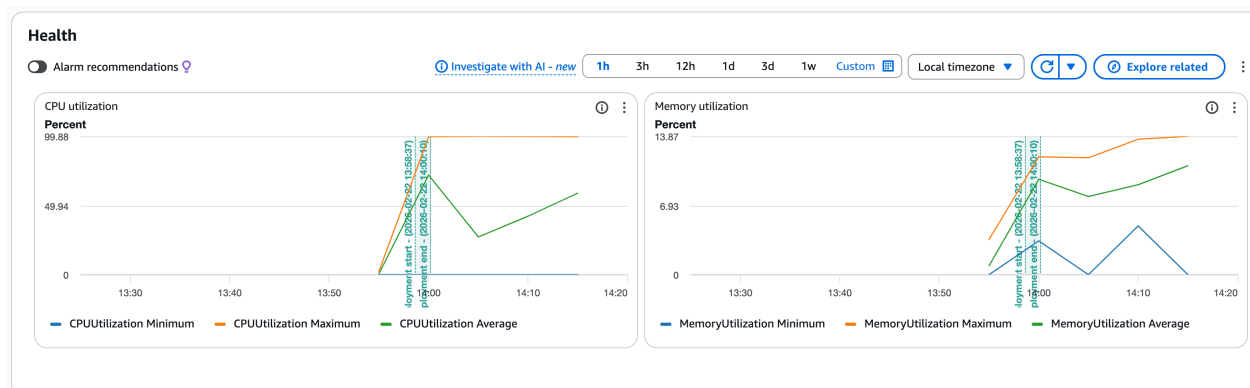
DOWNLOAD DATA

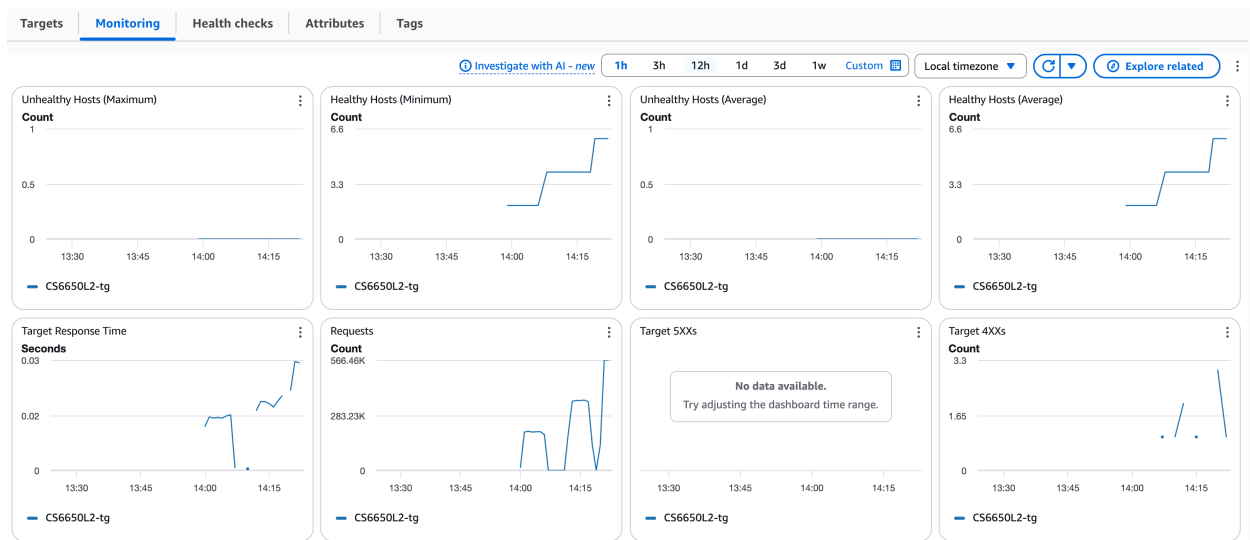
LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/products/search?q=alpha	661957	0	45	140	200	63.42	24	1227	1094.95	508.7	0
GET	/products/search?q=beta	662548	0	45	140	200	63.39	23	1214	1044.95	499.9	0
GET	/products/search?q=books	661607	0	45	140	200	63.39	23	1231	1044.95	505.5	0
GET	/products/search?q=electronics	660726	0	45	140	200	63.35	24	1212	1094.94	516.9	0
GET	/products/search?q=home	661706	0	45	140	200	63.28	24	1202	974.95	511.5	0
GET	/products/search?q=product	662512	0	45	140	200	63.41	24	1225	1971.94	517.1	0
	Aggregated	3971056	0	45	140	200	63.37	23	1231	1204.59	3059.6	0



400 users:





 LOCUST

Host

http://CS6650L2-alb-1624841869.us-east-1.elb.ama...

Status

STOPPED

RPS

6100.9

Failures

0%

NEW

RESET



STATISTICS

CHARTS

FAILURES

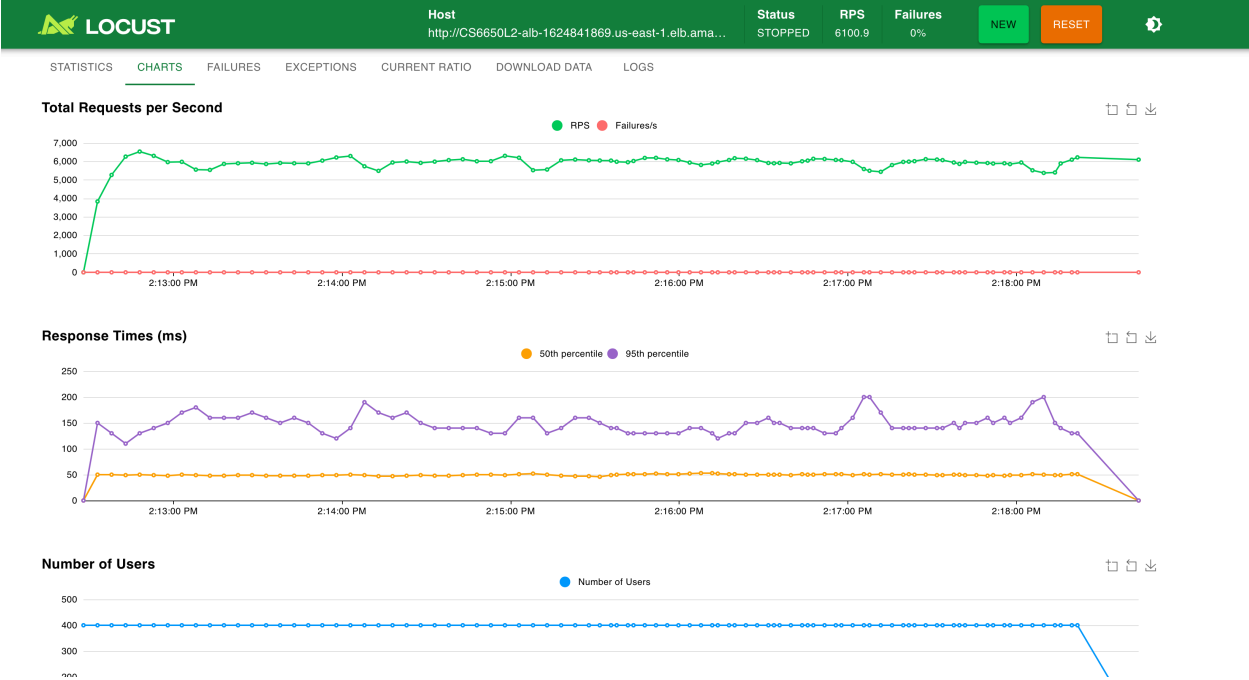
EXCEPTIONS

CURRENT RATIO

DOWNLOAD DATA

LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/products/search?q=alpha	351994	0	50	140	240	65.7	27	761	1094.92	1027.6	0
GET	/products/search?q=beta	351513	0	50	140	240	65.68	24	747	1044.92	1012.6	0
GET	/products/search?q=books	350815	0	50	140	240	65.61	24	612	1044.92	1019.9	0
GET	/products/search?q=electronics	351725	0	50	150	240	65.74	26	755	1094.92	1015.7	0
GET	/products/search?q=home	351402	0	50	140	240	65.53	26	612	974.92	1019	0
GET	/products/search?q=product	350662	0	50	150	240	65.86	25	744	1971.92	1006.1	0
	Aggregated	2108111	0	50	140	240	65.69	24	761	1204.14	6100.9	0



800 users:

LOCUST Host: http://CS6650L2-alb-1624841869.us-east-1.elb.ama... Status: STOPPED RPS: 8832.7 Failures: 0% NEW RESET

STATISTICS CHARTS FAILURES EXCEPTIONS CURRENT RATIO DOWNLOAD DATA LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/products/search?q=alpha	599149	0	54	220	350	80.57	27	1112	1094.92	1498.6	0
GET	/products/search?q=beta	598738	0	54	220	350	80.52	27	1033	1044.92	1459.1	0
GET	/products/search?q=books	599675	0	54	220	340	80.42	26	819	1044.92	1476.9	0
GET	/products/search?q=electronics	601011	0	54	220	350	80.51	27	849	1094.92	1459.6	0
GET	/products/search?q=home	598893	0	54	220	340	80.43	26	839	974.92	1461.5	0
GET	/products/search?q=product	599664	0	54	220	350	80.49	28	976	1971.91	1477	0
Aggregated		3597130	0	54	220	350	80.49	26	1112	1204.48	8832.7	0

